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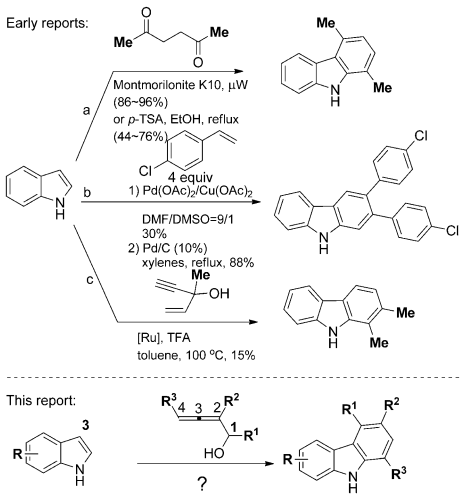
Carbazoles from the [4C + 2C] Reaction of 2,3-Allenols with Indoles

Binjie Guo, Xin Huang, Chunling Fu, and Shengming Ma*[a]

Abstract: A mild and efficient method using readily available 1-aryl-2,3-allenols and unprotected-N indoles, Au^+-catalyzed cyclization, and aromatization to afford the final [4C + 2C] products, carbazoles **4,** with an excellent selectivity, is reported. The reaction demonstrates excellent regioselectivity and allows the N–H unit to undergo reactivity unprotected. A mechanism involving a spiropolycyclic intermediate has been proposed and synthetic application is also demonstrated.

Tricyclic carbazole represents a key molecular motif that is widespread in plants, and many of these compounds show biological activities such as anti-oxidative, anti-bacterial, anti-microbial, psychotropic, anti-histaminic, anti-inflammatory, and antibiotic.[1] At the same time, carbazole derivatives are also widely used as organic materials with special thermal,[2] electrical,[3] optical,[4] electroluminescent,[5] hole-transporting, and light-emitting properties.[6] The important potential of these carbazole derivatives has made them attractive targets for organic synthesis.[7] Although there are a number of useful synthetic methods for the preparation of carbazoles,[8–10] there remain some limitations such as low yields and regioselectivity, poor atom economy, and harsh reaction conditions. These methods mostly involve N-protected indoles, which require initial protection and deprotection of the N-substituent. In fact, to the best of our knowledge, there are only very few records describing the construction of N–H carbazole skeletons directly from the N–H indole with no chemical pre-modification to the N–H (Scheme 1). Such examples include: a) the reactions of indoles with symmetric hexan-2,5-dione catalyzed by montmorilonite K10 under microwave conditions[11a] or *p*-toluenesulfonic acid (*p*-TSA) under reflux provide carbazoles in yields of 86–96% or 44–76%, respectively.[11b] Here, there remains an issue of regioselectivity when unsymmetric 1,4-ketones are applied. Other reports of N–H-carbazole synthesis are: b) the reaction of indole with two molecules of the same 1-chloro-4-vinylbenzene under Pd/Cu catalysis followed by 6e-cyclization affords 2,3-bis(4-chlorophenyl)-9*H*-carbazole in a 26% combined yield;[12] and c) the reaction of indole with 3-methylpent-1-en-4-yn-3-ol catalyzed by a Ru complex produces 1,2-dimethyl-9*H*-carbazole but in very low yield (15%), and also with an issue of regioselectivity.[13] In addition, Liang et al. reported an isolated example of 1-indol-3-yl-3-alkynol with ICl affording non-N-H 3-iodocarbzaole.[14] We envisioned to use differently substituted 2,3-allenols as the [4C] unit to react with indoles,[15] which may provide a straightforward route to carbazoles with a very decent diversity and regioselectiovity by considering the unique C3 reactivity of indoles towards the C1 position of 2,3-allenols. Here, we wish to report the realization of such a concept with the unprotected indoles[16] (Scheme 1).

We observed that the reaction of N–H indole **2a** with 1-(*p*-methoxy)phenyl-substituted 2,3-allenol **1a** catalyzed by $TsOH{\cdot}H_2O$ (5 mol%) afforded the 3-allenylated product **3a** (Table 1, entry 6). Other acids have also been screened, but better results were not obtained (Table 1, entries 1–5). Reducing the loading of the catalyst led to a lower yield of **3a** and a recovery of the starting material **1a** (Table 1, entries 7–9).



Scheme 1. Approaches to carbazoles from indoles with an N–H unit (TFA = trifluoroacetic acid).

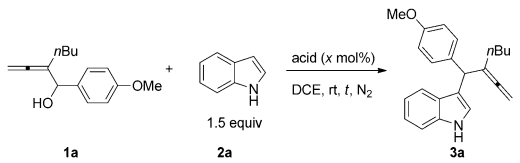
As we know, gold catalysis has been applied to the non-N–H-carbazole synthesis by cyclization and water-elimination of differently substituted indolyl-substituted propargylic alcohols,[17] and in the reaction of 1-indolyl-substituted 2,3-allenols via a gold carbene intermediate.[18,19] Although no cyclization occurred with $AuCl(PPh_3)$ in our system (Table 2, entry 1), we

[a] Dr. B. Guo, Dr. X. Huang, Prof. Dr. C. Fu, Prof. Dr. S. Ma
Laboratory of Molecular Recognition and Synthesis
Department of Chemistry, Zhejiang University
Hangzhou, 310027, Zhejiang (P.R. China)
E-mail: masm@mail.sioc.ac.cn

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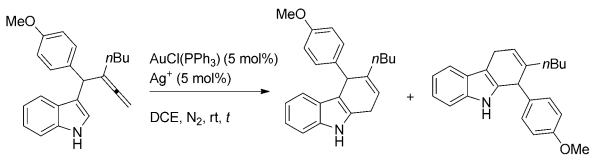
Table 1. Optimization of the reaction of indole **2a** with allenol **1a**.[a]



Entry	x (mol%)	t [h]	Yield of 3a [%][b]	Recovery of 1a [%][b]
1	HOAc (5)	28.5	0	87
2	benzoic acid (5)	23	0	92
3	L-tartaric acid (5)	42	0	87
4	CF_3COOH (5)	23.5	64	4
5	(*R*)-BNPPA[c] (5)	12.5	71	0
6	**TsOH·H_2O (5)**	**12.5**	**70 (74)**	**0**
7	TsOH·H_2O (2)	24.5	44	39
8	TsOH·H_2O (3)	24	68	15
9	TsOH·H_2O (4)	28	72	2

[a] All reactions were performed with **1a** (0.2 mmol), **2a** (0.3 mmol), and acid in 1 mL of DCE unless otherwise noted. [b] ^{1}H NMR yield of the crude product using CH_2Br_2 as the internal standard, and the isolated yield is given in parentheses. [c] (*R*)-BNPPA = (*R*)-(−)-1,1′-binaphthyl-2,2′-diyl hydrogen phosphate.

Table 2. Optimization of the cyclization reaction of **3a**.[a]



Entry	Ag^+	t [h]	Yield of 4a [%][b]	Yield of 4a′ [%][b]
1[c]	–	48	0	0
2	$AgSbF_6$	42	72	9
3	$AgBF_4$	12.5	82 (85)	4
4	$AgPF_6$	17.5	75 (79)	12 (9)
5	$AgNO_3$	18	32	0
6[d]	AgOAc	12.5	4	9
7	$AgNTf_2$	18	78	8
8	AgOTf	4.7	84	5
9	$AgBF_4$[e]	12.5	66	16
10	$AgBF_4$[f]	12.3	71	9
11	$PtCl_2$[g]	48	21	13
12[h]	$AgBF_4$	13	78	0
13[i]	$AgBF_4$	18	69	0

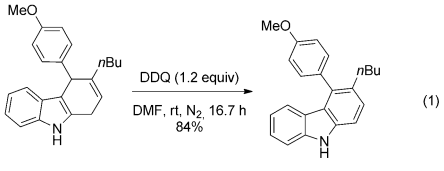
[a] All reactions were performed with **3a** (0.2 mmol), Au^+ (5 mol%), and Ag^+ (5 mol%) in 1 mL of DCE at rt unless otherwise noted. [b] ^{1}H NMR yield of the crude product using CH_2Br_2 as the internal standard, and the isolated yield is given the parentheses. [c] 79% of **3a** was recovered. [d] 72% of **3a** was recovered. [e] $AuCl_3$ was used instead of $AuCl(PPh_3)$. [f] AuCl was used instead of $AuCl(PPh_3)$. [g] $PtCl_2$ was used instead of Au^+/Ag^+ as catalyst. [h] The reaction was conducted at 0 °C. [i] The reaction was conducted at −10 °C.

were delighted to see that this Au catalyst and $AgSbF_6$ could be used as co-catalysts and the expected tricyclic product **4a** could be obtained in a yield of 72% along with the isomer **4a**′ (9%; Table 2, entry 2). Screening of other Ag^I salts led to the observation that $AgBF_4$ was the best co-catalyst in terms of yield and regioselectivity (Table 2, entries 2–8). $AuCl_3$ and AuCl were also tested, but no better result was obtained (Table 2, entries 9 and 10). Using $PtCl_2$ in place of the Au/Ag system also gave a poor result (Table 2, entry 11). To our delight, when the reaction was conducted at 0 °C, product **4a** was exclusively formed in a yield of 78% (Table 2, entry 12). Performing the reaction at −10 °C gave a lower yield of 69% (Table 2, entry 13).

Subsequent treatment of the crude **4a** with 2,3-dichloro-5,6-dicyano-*p*-benzoquinone (DDQ) in DMF at room temperature afforded **5a** in a yield of 84% [Eq. (1)].

The regioselectivity of the current protocol was established by the X-ray diffraction study of **5a**[20] (Figure 1).

With the optimized reaction conditions in hand, we next investigated the scope of substrates. The reactions of allenols with different R groups at the 2-position such as), *n*-butyl, methyl, *n*-propyl, 2-(2-phenyl)ethyl, 2-(3-phenyl)propyl, isopropyl, cyclohexyl, 2-allyl, 2-benzyl, and phenyl (**1a–j**, respectively) occurred smoothly affording products **3** (approx. 65–90%) and **5** (approx. 38–86%) in moderate to good yields (Table 3).



(1)

Moreover, other aryl groups at the 1-position of the allenols were also accommodated: 4-ethoxyphenyl- (**1k**), 1-(benzo[*d*][1,3]dioxol-5-yl)- (**1l** and **1m**) and 3,4,5-trimethoxyphenyl-substituted (**1n**) allenols all reacted smoothly under the optimized reaction conditions (Table 4, entries 1–4). When heteroaryl 2-

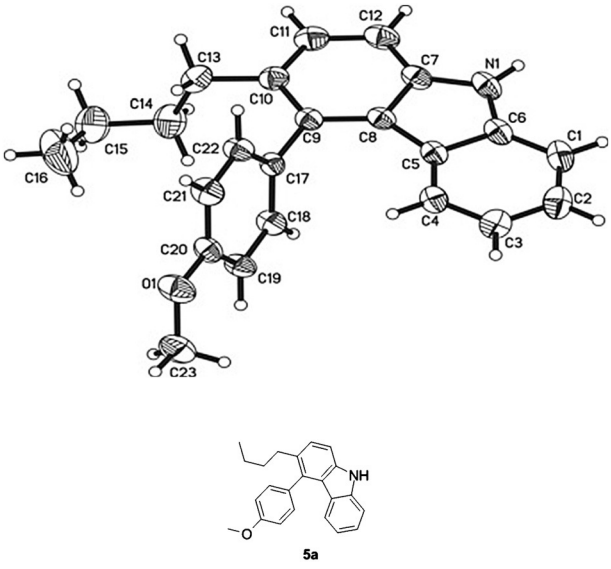


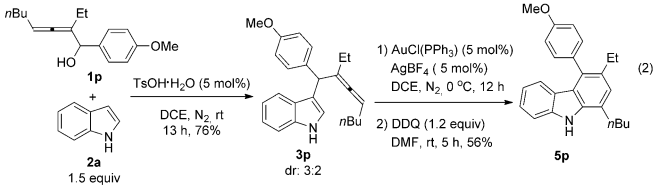
Figure 1. X-ray diffraction study of **5a**.

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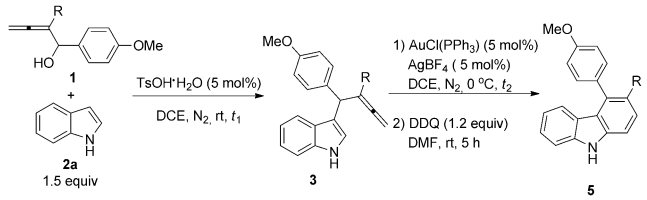
thienyl (**1o**) was introduced, similar results were observed (Table 4, entry 5).

Furthermore, the reaction of **1p** bearing both a butyl group at the 4-position and ethyl group at the 2-position with **2a** under the optimized reaction conditions afforded the expected poly-substituted indolylallene **3p** (76%) and carbazole **5p** (56%). [Eq. (2)]



Indoles with different substitutions were also explored (Table 5). Indoles with a 5-, 6-, or 7-methyl substituent all reacted with **1c** affording products **3q**–**s** and **5q**–**s** in moderate to good yields (Table 5, entries 1–3). 5-Methoxyindole (**2e**) gave **3t** in a yield of 69% and **5t** in 29% yield (Table 5, entry 4), whereas 5-methoxycarbonyl-substituted indole (**2f**) gave **3u** in a yield of 53% and **5u** in a yield of 34% (Table 5, entry 5);

Table 3. Substrate scope of allenols for the formation of carbazoles.[a]



Entry	R	t_1 [h]	Yield of **3** [%][b]	t_2 [h]	Yield of **5** [%][b]
1	*n*Bu (**1a**)	18	72 (**3a**)	13.2	74 (**5a**)
2	Me (**1b**)	11.7	65 (**3b**)	15.5	71 (**5b**)
3[c]	*n*Pr (**1c**)	13.5	77 (**3c**)	8.5	67 (**5c**)
4	$(CH_2)_2$Ph (**1d**)	23	76 (**3d**)	11.3	79 (**5d**)
5	$(CH_2)_3$Ph (**1e**)	16.5	74 (**3e**)	18.7	68 (**5e**)
6	*i*Pr (**1f**)	13	80 (**3f**)	11.5	38 (**5f**)
7	cyclohexyl (**1g**)	14	90 (**3g**)	12	69 (**5g**)
8	allyl (**1h**)	16	76 (**3h**)	11.5	65 (**5h**)
9[d]	Bn (**1i**)	13	62 (**3i**)	12	86 (**5i**)
10	Ph (**1j**)	24	59 (**3j**)	12	54 (**5j**)

[a] All reactions were performed with **1** (2 mmol), **2a** (3 mmol), and TsOH·H_2O (5 mol%) in 10 mL of DCE affording **3**. Then the reaction of **3** (1 mmol), AuCl(PPh_3) (5 mol%), and $AgBF_4$ (5 mol%) in 5 mL of DCE gave a crude product, which was then reacted with DDQ (1.2 equiv) in 5 mL of DMF to afford product **5**. [b] Isolated yield. [c] The crude **4c** was treated with DDQ for 12.5 h. [d] The crude **4i** was treated with DDQ for 4 h.

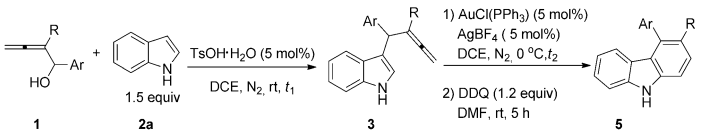
Both 5- and 6-bromoindoles worked smoothly with **1c** to give the products **3v**,**w** and **5v**,**w** in good yields (Table 5, entries 6 and 7).

To demonstrate the practicality of this method, a gram scale reaction of **1a** (1.3921 g, 6 mmol) with **2a** (1.0539 g, 9 mmol) was conducted to afford the corresponding product **3a** (1.4958 g) in a yield of 75%. Subsequent cyclization and aromatization of **3a** (1.3231 g, 4 mmol) with DDQ (1.0895 g, 4.8 mmol) afforded the corresponding product **5a** (1.0039 g) with a yield of 76% [Eq. (3)]. Furthermore, such a reaction may also be conducted without purification affording **5a** in a 47% combined yield [Eq. (4)].

These alkoxy groups on the aryl ring allow further opportunity for subsequent synthetic elaboration. Further treatment of **5a** with BBr_3 in CH_2Cl_2 followed by esterification with trifluoromethanesulfonic anhydride (Tf_2O) afforded **5ab** in 84% yield (two steps, Scheme 2). In the presence of 5 mol% of $Pd(OAc)_2$, 5 mol% of LB-Phos·HBF_4[21] and 3 equiv of K_3PO_4·3H_2O, **5ab** was successfully coupled with phenylboronic acid to give the product **5ac** in 94% yield. Similarly, **5ad** was obtained in a yield of 81% by the coupling reaction of **5ab** with (*E*)-styrylboronic acid in the presence of 5 mol% of $Pd(PPh_3)_4$ and 3 equiv of K_3PO_4·3H_2O. In addition, the reaction of **5ab** with phenylacetylene co-catalyzed by 5 mol% each of $Pd(PPh_3)_4$ and AuCl(PPh_3)[22] also proceeded smoothly to afford **5ae** in 93% yield (Scheme 2).

A possible mechanism that explains the formation of **4a** and **4a′** is illustrated in Scheme 3. The reaction of allenol **1a** with TsOH generated carbocation **IM1**. Subsequent electrophilic substitution of indolyl C3 with the allenic carbocation afforded **3a** along with the regeneration of TsOH. Then the coordination of Au^+ to the terminal C=C of the allene fol-

Table 4. Substrate scope of allenols for the formation of carbazoles.[a]



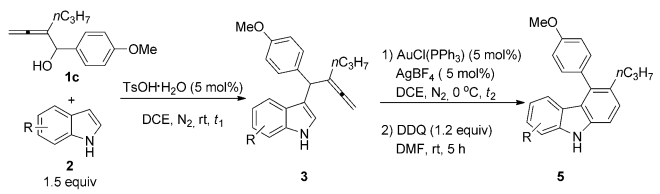
Entry	Ar	R	t_1 [h]	Yield of **3** [%][b]	t_2 [h]	Yield of **5** [%][b]
1	4-$EtOC_6H_4$	*n*Bu (**1k**)	16.5	61 (**3k**)	16	72 (**5k**)
2[c]	3,4-$OCH_2OC_6H_3$	*n*Bu (**1l**)	12	70 (**3l**)	12	78 (**5l**)
3[c]	3,4-$OCH_2OC_6H_3$	Ph (**1m**)	24	45 (**3m**)	11.5	79 (**5m**)
4[d]	3,4,5-$(MeO)_3C_6H_2$	Et (**1n**)	130	68 (**3n**)	11.5	53 (**5n**)
5	2-thienyl	*n*Bu (**1o**)	11.7	64 (**3o**)	12	42 (**5o**)

[a] All reactions were performed with **1** (2 mmol), **2a** (3 mmol), and TsOH·H_2O (5 mol%) in 10 mL of DCE to give **3**. Then the reaction of **3** (1 mmol), AuCl(PPh_3) (5 mol%), and $AgBF_4$ (5 mol%) in 5 mL of DCE gave a crude product, which was then reacted with DDQ (1.2 equiv) in 5 mL of DMF to afford product **5**. [b] Isolated yield. [c] The crude **4** reacted with DDQ for 4.3 h. [d] The reaction of **1n** with **2a** was carried out at 50 °C.

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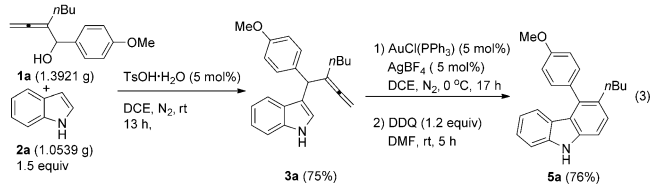
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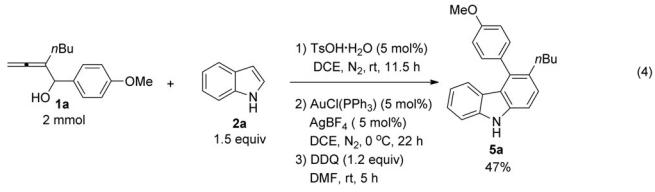
Table 5. Substrate scope of indoles for the formation of carbazoles.[a]



Entry	R	t_1 [h]	Yield of **3** [%][b]	t_2 [h]	Yield of **5** [%][b]
1	5-Me (**2b**)	13	74 (**3q**)	12.8	51 (**5q**)
2[c]	6-Me (**2c**)	19	59 (**3r**)	23.5	62 (**5r**)
3	7-Me (**2d**)	16.6	76 (**3s**)	11.5	80 (**5s**)
4	5-MeO (**2e**)	7.2	69 (**3t**)	11	29 (**5t**)
5[d]	5-MeO_2C (**2f**)	24	53 (**3u**)	12	34 (**5u**)
6	5-Br (**2g**)	12	66 (**3v**)	10.5	66 (**5v**)
7	6-Br (**2h**)	24	72 (**3w**)	11	75 (**5w**)

[a] All reactions were performed with **1c** (2 mmol), 2 (3 mmol), and $TsOH·H_2O$ (5 mol%) in 10 mL of DCE to give **3**. Then the reaction of **3** (1 mmol), $AuCl(PPh_3)$ (5 mol%), and $AgBF_4$ (5 mol%) in 5 mL of DCE gave a crude product, which was then reacted with DDQ (1.2 equiv) in 5 mL of DMF to afford product **5**. [b] Isolated yield. [c] The crude **4r** reacted with DDQ for 5.5 h. [d] The reaction of **1c** with **2f** was carried out at 50 °C.





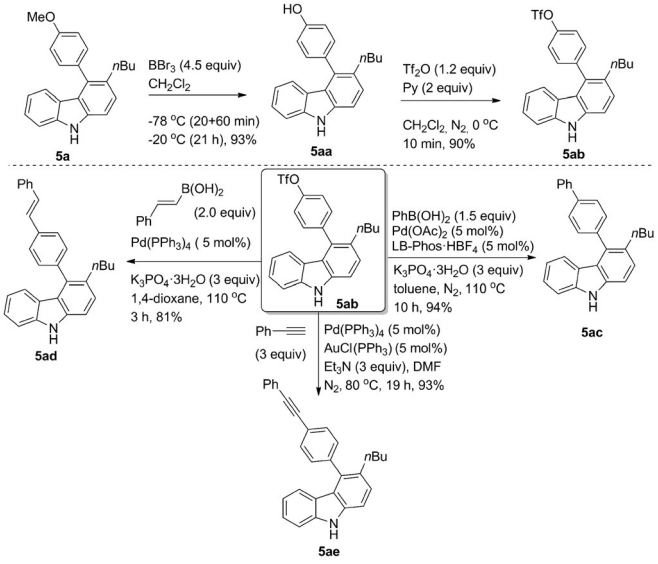
lowed by intramolecular attack of the indolyl C3 furnished spirocyclic species **IM2**, based on the detection of both **4a**/**4a′** (see Table 2).[23,24] Subsequent 1,2-alkyl migration/proto-demetalation sequence affords dihydrocarbazole **4a** (path a) and **4a′** (path b). By fine tuning the reaction parameters, excellent selectivity of **4** and **4′** has been achieved.

In conclusion, we have developed a very mild and efficient method using readily available 1-aryl-2,3-allenols[25] and *N*-non-protected indoles by a Friedel–Crafts-type C–C bond formation, Au^+/Ag^+-catalyzed cyclization, 1,2-carbon shift, and aromatization to afford the final [4C + 2C]-products, N–H carbazoles. The reaction demonstrates excellent regioselectivity through the installation of specific substituents at the different locations of the newly formed benzene ring in the indoles by utilizing the substituent-loading capability of 1-(electron-rich aryl)-2,3-allenols. The unprotected nature of the N–H unit leaves further opportunity for further purpose-directed modification. The application of the method for organic synthesis, materials science, and bio-screening are being carried out in our laboratory.

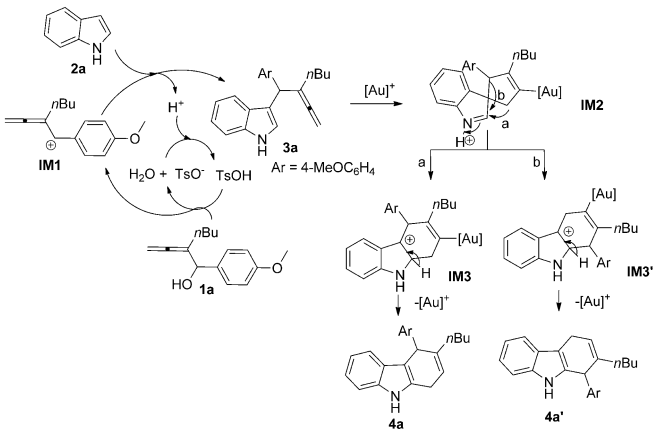
Experimental Section

Preparation of 3a

To a dried Schlenk vessel was added allenol **1a** (1.3921 g, 6 mmol) and DCE (1,2-dichloroethane; 30 mL) under N_2 atmosphere at room temperature, followed by the addition of indole **2a** (1.0539 g, 9 mmol) and $TsOH·H_2O$ (57.2 mg, 0.3 mmol). The resulting mixture was stirred at room temperature until the reaction was completed as monitored by TLC after 13 hours. The reaction mixture was filtered through a short column of silica gel (ethyl ether, 10 mL×2). Evaporation and purification by chromatography (petroleum ether/ethyl acetate = 50:1 to 30:1 to 20:1) on silica gel afforded **3a** (1.4958 g, 75%): Oil; ^1H NMR (300 MHz, $CDCl_3$): $\delta = 7.87$ (s, 1H, NH), 7.49 (d, $J = 7.8$ Hz, 1H, ArH), 7.30 (td, $J = 8.1$, 1.1 Hz, 1H, ArH), 7.27–7.19 (m, 2H, ArH), 7.15 (t, $J = 7.5$ Hz, 1H, ArH), 7.05 (t, $J = 7.4$ Hz, 1H, ArH), 6.83 (d,



Scheme 2. Application of compound **5a**.



Scheme 3. A proposed mechanism for the formation of **4a** and **4a′**.

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J=8.7 Hz, 2H, ArH), 6.73 (dd, J=2.4, 0.9 Hz, 1H, ArH), 4.75 (s, 1H, CH), 4.60–4.45 (m, 2H, $=CH_2$), 3.78 (s, 3H, OCH_3), 2.08–1.97 (m, 2H, CH_2), 1.54–1.41 (m, 2H, CH_2), 1.40–1.24 (m, 2H, CH_2), 0.87 ppm (t, J=7.2 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ=207.1, 157.9, 136.4, 135.0, 129.6, 127.2, 122.9, 121.8, 119.7, 119.2, 118.8, 113.3, 110.9, 107.2, 77.2, 55.2, 45.6, 31.3, 30.0, 22.5, 14.0 ppm; IR (neat) $\tilde{\nu}$=3419, 3057, 2997, 2953, 2928, 2872, 2857, 2834, 1955, 1610, 1583, 1506, 1456, 1417, 1337, 1301, 1246, 1175, 1094, 1035, 1011 cm^{-1}; MS (70 ev, EI) m/z (%) 331 (M^+, 11.96), 236 (100); HRMS calcd for $C_{23}H_{25}NO$ (M^+): 331.1936; found: 331.1941.

Preparation of 5a

To a dried Schlenk vessel was added **3a** (1.3231 g, 4 mmol) and DCE (20 mL). After stirring at 0 °C for 10 minutes, $AuCl(PPh_3)$ (99.1 mg, 0.2 mmol) and $AgBF_4$ (39.2 mg, 0.2 mmol) were added. The resulting mixture was stirred at 0 °C until the reaction was completed as monitored by TLC after 17 h. The reaction mixture was filtered through a short column of silica gel (diethyl ether, 10 mL×2) and evaporated. DDQ (1.0895 g, 4.8 mmol) was added to the solution of the crude product in DMF (20 mL) and stirred at room temperature for 5 hours. Then 10 mL of water was added to quench the reaction. The resulting mixture was extracted with diethyl ether (3×15 mL), washed with brine, and dried over anhydrous Na_2SO_4. Filtration, evaporation, and chromatography on silica gel (distilled petroleum ether (30≈60 °C)/ethyl acetate=30:1 to 20:1) of the crude product afforded **5a** (1.0039 g, 76%): Solid, m.p. 124.7–125.7 °C (*n*-hexane/CH_2Cl_2); ^{1}H NMR (300 MHz, $CDCl_3$): δ= 7.88 (s, 1H, NH), 7.35–7.18 (m, 6H, ArH), 7.05 (d, J=8.1 Hz, 2H, ArH), 6.93–6.82 (m, 1H, ArH), 6.75 (d, J=8.1 Hz, 1H, ArH), 3.91 (s, 3H, OCH_3), 2.55 (t, J=7.8 Hz, 2H, CH_2), 1.58–1.40 (m, 2H, CH_2), 1.32–1.15 (m, 2H, CH_2), 0.80 ppm (t, J=7.2 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ=158.8, 139.9, 137.6, 135.6, 132.5, 132.3, 130.5, 127.0, 125.1, 123.6, 122.4, 122.2, 118.9, 114.0, 110.2, 109.4, 55.3, 34.6, 32.2, 22.5, 13.9 ppm; IR (KBr) $\tilde{\nu}$=3404, 3057, 3032, 2994, 2955, 2926, 2857, 2834, 2533, 1610, 1515, 1490, 1451, 1382, 1333, 1283, 1244, 1175, 1151, 1127, 1105, 1087, 1033 cm^{-1}; MS (70 ev, EI) m/z (%) 329 (M^+, 53.23), 286 (100); elemental analysis calcd for $C_{23}H_{23}NO$: C 83.85, H 7.04, N 4.25; found: C 83.64, H 7.10, N 4.09.

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Keywords: allenes · aromatization · carbazoles · cyclization · indoles

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